CPI (Consumer Price Index)

- Measures how much everyday prices are going up for common goods and services.
- For our project,

WTI (West Texas Intermediate) → Oil

- The main crude oil (unrefined petroleum product) price used in the U.S.
- WTI was used in our project mainly because it's the most relevant benchmark for U.S. domestic energy pricing.
    - **Brent** and **Natural Gas** were used mainly for comparison (to see whether different energy types behaved differently in relation to inflation)

MMBtu → A standard unit of energy representing 1 million British Thermal Units (Btu), used primarily to measure and price natural gas and other energy sources.

Brent (Crude Oil)

- The main oil price used internationally
- It's tracked closely with WTI (the correlation between the two was r = 0.98, meaning they moved almost identically)

Natural Gas

- The price of gas used for energy, such as heating and electricity

Tanker Arrivals

- How many oil ships are coming into the Strait of Hormuz each month (IMF Portwatch)
    - **Tankers**: vessels that carry crude oil
- Tanker arrivals sort of measure how much oil is physically moving through the Strait

---

Goal

The goal of this project can be divided into three

1. How inflation and energy prices move together over time
2. Whether that relationship changes during global crises
3. And the tanker traffic(Strait of Hormuz) is used to interpret those factors in a supplementary manner. (In our report, we didn't write that it is a supplement, but it is actually a supplementary factor.)

Datasets

We have three initial datasets:

1. Arrivials-of-ships.csv → daily tanker and vessel arrivals at the Strait of Hormuz
2. USCPI2.csv → Monthly U.S. CPI-U index values from 1913 to July 2021
3. Resources_Dataset.csv → Daily commodity prices including WTI, Brent, and Natural Gas from 1997 to August 2025

Note:

Joan cleaned those three datasets in notebook1, notebook2, and notebook3.

The cleaned datasets are:

1. hormuz_arrivals_monthly.csv
2. cpi_inflation_monthly.csv
3. Commoditiy_prices_monthly.csv

In the analysis notebook, those three cleaned datasets are called and converted to dataframes.

Because the datatype of each "YearMonth" column is string, the changing type to date_time was computed.

After that, the "YearMonth" of each dataset was set as an index, and it was sorted again just in case.

Merging data

CPI rows: 19 | Range: 2020-01-31 -> 2021-07-31

Resources rows: 72 | Range: 2020-01-31 -> 2025-12-31

Hormuz rows: 72 | Range: 2020-01-31 -> 2025-12-31

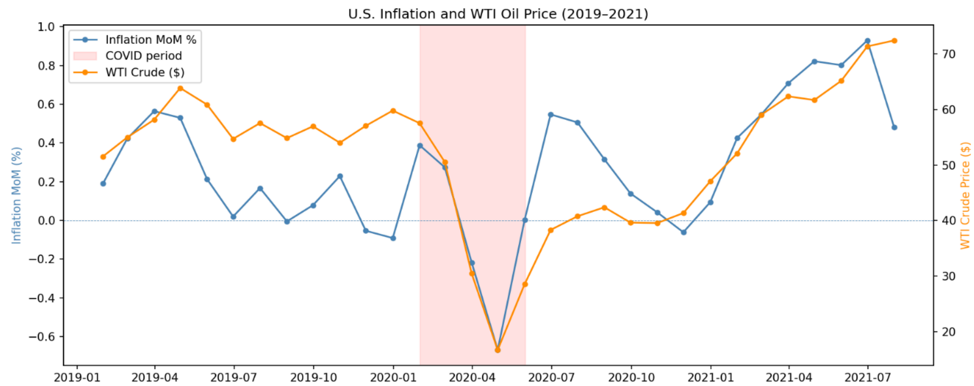
Because the period of CPI we have is different from the other 2 datasets, we need to merge in two ways.

All-three window: 2020-02-29 -> 2021-07-31 | 18 months

Resources+Hormuz: 2020-01-31 -> 2025-12-31 | 72 months

Visualizations

Visualization 1 → Inflation and WTI (Oil) Price Over Time



Goal: To see if *inflation* (blue) and *oil* (yellow) move together across that 2019-2021 window

- (Smaller period in summary stats, but this shows the full story)

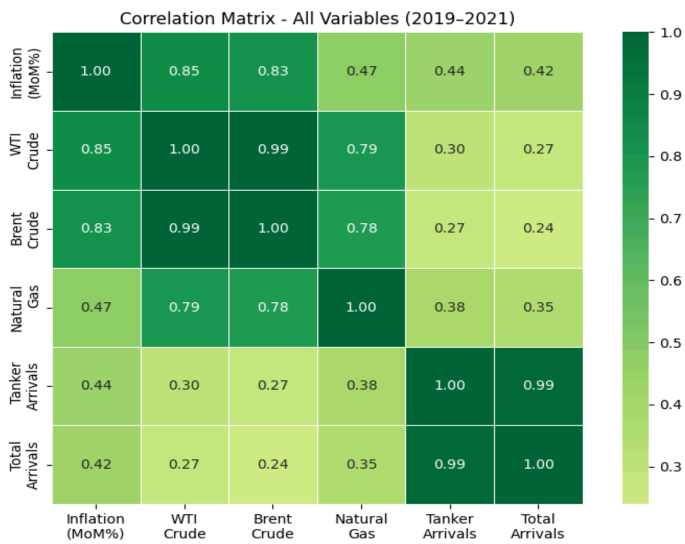
Result: They mostly moved together, especially during COVID period in early 2020, when they dropped significantly and then recovered together.

- I used two y-axes (one for inflation of the left and one for oil on the right)
    - This method was used so metrics could be shown on the same timeline despite having different units of measurement (Inflation → Percentage; Oil Price → $).

The reason why this graph is chosen for the final report:

- This visualization was included because it shows the core idea of the project.
    - During the COVID period, both inflation and oil prices dropped significantly, then recovered.
    - This suggests that they tend to move together, at least during certain periods.

Visualization 2 → Pearson Correlation Heatmap



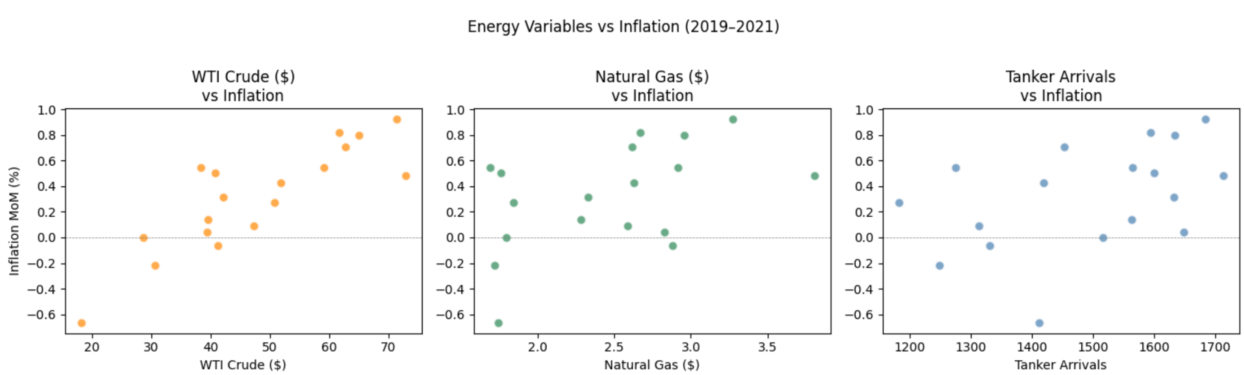
	Inflation (MoM%)	WTI Crude	Brent Crude	Natural Gas	Tanker Arrivals	Total Arrivals
Inflation (MoM%)	1.00	0.85	0.83	0.47	0.44	0.42
WTI Crude	0.85	1.00	0.99	0.79	0.30	0.27
Brent Crude	0.83	0.99	1.00	0.78	0.27	0.24
Natural Gas	0.47	0.79	0.78	1.00	0.38	0.35
Tanker Arrivals	0.44	0.30	0.27	0.38	1.00	0.99
Total Arrivals	0.42	0.27	0.24	0.35	0.99	1.00

What is a correlation heatmap?

A Pearson correlation heatmap is a grid-style chart that displays the Pearson correlation coefficient **(r)** between every possible pair of variables in a dataset simultaneously. Instead of running separate comparisons one at a time, the heatmap shows all relationships at once, with color coding to make the strongest and weakest ones immediately visible.

- **(r)** ⟶ stands for the Pearson correlation coefficient. It is a single number between −1 and +1 that summarizes the strength of the linear relationship between two variables.
- How to interpret the strength of each relationship:
  - **0.7 to 1.0** ⟶ Strong relationship. The two variables move together closely and consistently.
  - **0.4 to 0.7** ⟶ Moderate relationship. There is a real pattern, but it is not tight.
  - **0.1 to 0.4** ⟶ Weak relationship. A slight tendency exists, but it is easily disrupted.
  - **0.0 to 0.1** ⟶ Negligible. Essentially, no linear relationship.
- Looking at our heatmap, the top row is the most important:
  - It shows how strongly each variable correlates with inflation (MoM%)
  - **WTI** at r = 0.85 and **Brent** at r = 0.83 are in the **strong** range, meaning oil prices and inflation moved closely together in our 2020–2021 window
  - **Natural gas** at r = 0.47 falls in the **moderate** range
  - **Tanker** and **Total arrivals** at r = 0.44 and r = 0.42 are also **moderate**, but weaker than the oil prices.
  - The near-zero values between **Tanker** and **WTI**, r = 0.08, and **Tanker** and **Brent**, r = −0.00, are important too.
    - They show that in our short window, the number of ships passing through Hormuz had almost no direct relationship with oil prices

Visualization 3 ⟶ Scatter Plots (Each Variable v. Inflation)



Axes:

Y-Axis ⟶ Inflation

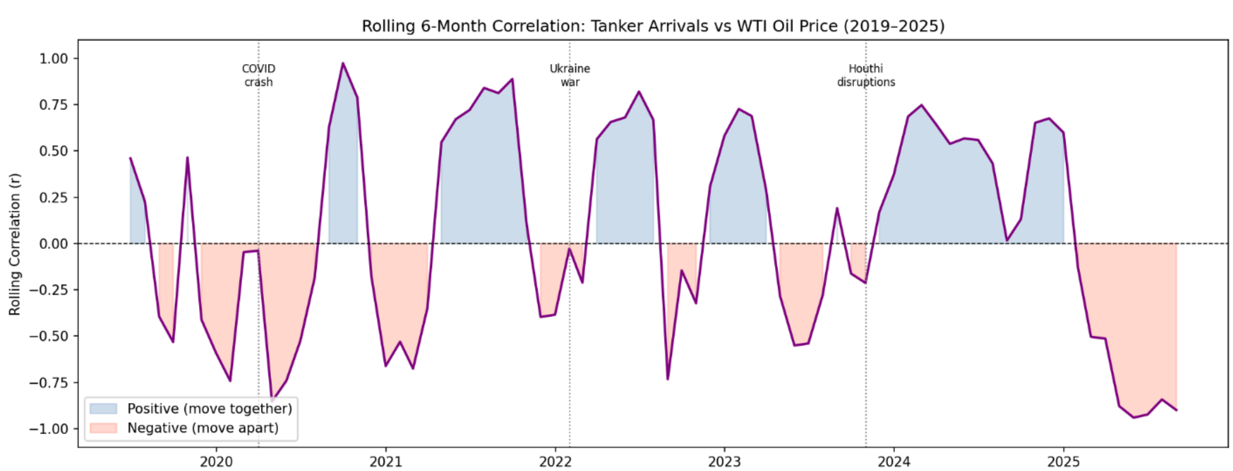
X-Axis ⟶ Value of oil, natural gas, and ship arrivals

- We're looking at how much consumer prices changed in a month
- Values above zero mean prices rose, values below zero mean prices fell.
  - The x-axis changes across the three panels:
    - WTI crude price in dollars for the first, Natural Gas price in dollars per MMBtu for the second, and monthly tanker arrival count at Hormuz for the third.
    - This targets the question ⟶ **Does a higher value of this energy variable correspond to higher inflation that month?**

How to Interpret This Graph:

- The WTI panel has the clearest diagonal tendency, Natural Gas is messier, and Tanker Arrivals shows the least directional structure.
  - Main takeaway: Oil prices are the strongest predictor of inflation in this window, and tanker traffic is the weakest
  - Notable outliers all have to do with the COVID period ⟶
    - WTI: The single dot in the far bottom-left, sitting around $17 on the x-axis and −0.65% on the y-axis, is the April 2020 month when WTI crashed, and inflation went deeply negative simultaneously
    - Natural Gas: The outlier dot in the bottom-left, around $1.50 and −0.65%, is the same April 2020 month appearing again
    - Tanker: Notable outlier sitting around 1,400 tankers and −0.65% inflation is again April 2020

Visualization 4 ⟶ Rolling 6-Month Correlation: Tanker v. WTI (Oil)



A rolling correlation chart shows how the relationship between two variables changes over time, instead of summarizing it as a single number. Rather than computing one r value across the entire dataset, a 6-month window slides forward one month at a time, recalculating r at each step. The result is a line that rises and falls, showing when the two variables were moving together and when they were not.

Axes:

- Y-Axis → Rolling Correlation (r), ranging from −1.0 to +1.0
  - Values above 0 mean tanker arrivals and WTI were moving in the same direction over the 6-month window
  - Values below 0 mean they were moving in opposite directions
- X-Axis → Time, from January 2019 to August 2025

What the colors mean:

- The **purple line** is the rolling correlation value itself
- **Blue shading** fills the area between the line and zero whenever the correlation is positive
  - These are periods where more tankers through Hormuz corresponded with higher oil prices, the expected normal pattern
- **Red shading** fills the area whenever the correlation is negative
  - These are periods where tanker activity and oil prices were moving in opposite directions, which is the anomalous pattern

What the vertical dotted lines mean: Three labeled event markers are drawn on the chart:

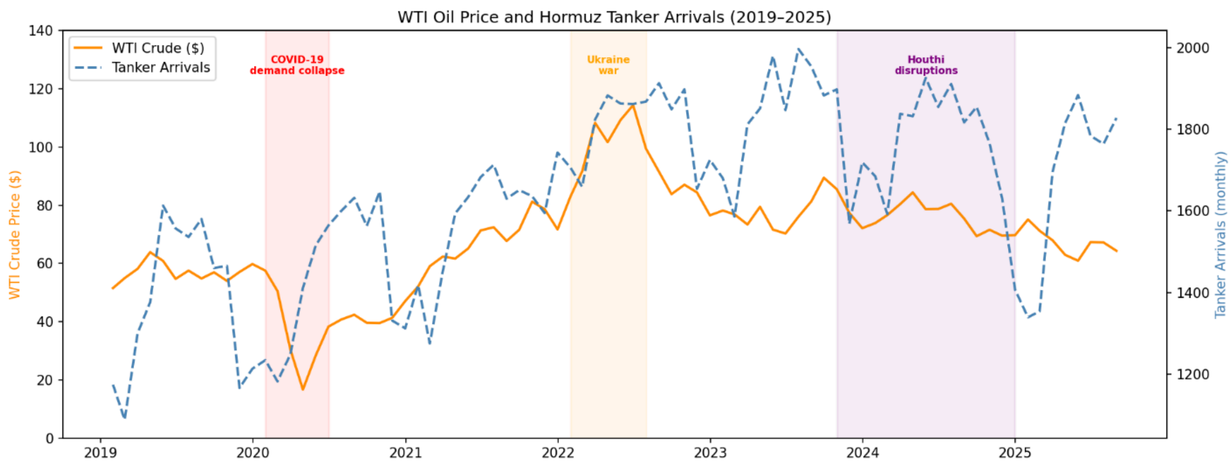
- **COVID crash (April 2020)**: when global oil demand collapsed, and both tanker activity and prices fell simultaneously
- **Ukraine war (February 2022):** when Russia invaded Ukraine, disrupting European energy supply chains and driving oil prices sharply higher
- **Houthi disruptions (November 2023)**: when Houthi attacks on Red Sea shipping began reducing tanker transits through Hormuz

How to interpret this chart:

- The line oscillating between blue and red throughout 2019–2022 reflects normal market volatility
  - The relationship between tankers and oil was unstable but not systematically negative
- The most important region is the **sustained red shading from late 2024 into 2025**, where the correlation drops to around −0.90. This means tanker arrivals were falling while oil prices remained elevated, a structural break from the usual pattern, directly caused by the Houthis' disruptions

- A single correlation number like r = 0.08 (from the heatmap) would completely hide this story. The rolling chart shows that the relationship is not stable. It flips the sign depending on what is happening geopolitically

Visualization 5 ⟶ WTI Oil Price + Tanker Arrivals with Event Markers



This is a dual-axis time series chart overlaying two variables on the same timeline, with shaded regions marking key geopolitical events. It is designed to let you see whether oil prices and tanker activity moved together or apart, and to connect those movements to real-world causes.

Axes:

- Left Y-Axis (orange) ⟶ WTI Crude Price in dollars per barrel, ranging from $0 to $140
- Right Y-Axis (blue) ⟶ Monthly Tanker Arrivals at the Strait of Hormuz, ranging from approximately 1,200 to 2,000 ships
- X-Axis ⟶ Time, from January 2019 to August 2025
    - Having two y-axes is necessary because the two variables use completely different units and scales (without this, one line would visually dominate the other)

What the lines mean:

- **Solid orange line** ⟶ WTI crude oil price each month
- **Dashed blue line** ⟶ Total tanker arrivals at Hormuz each month
    - The dashed style for tankers and the solid style for WTI make them easy to distinguish, even where they overlap

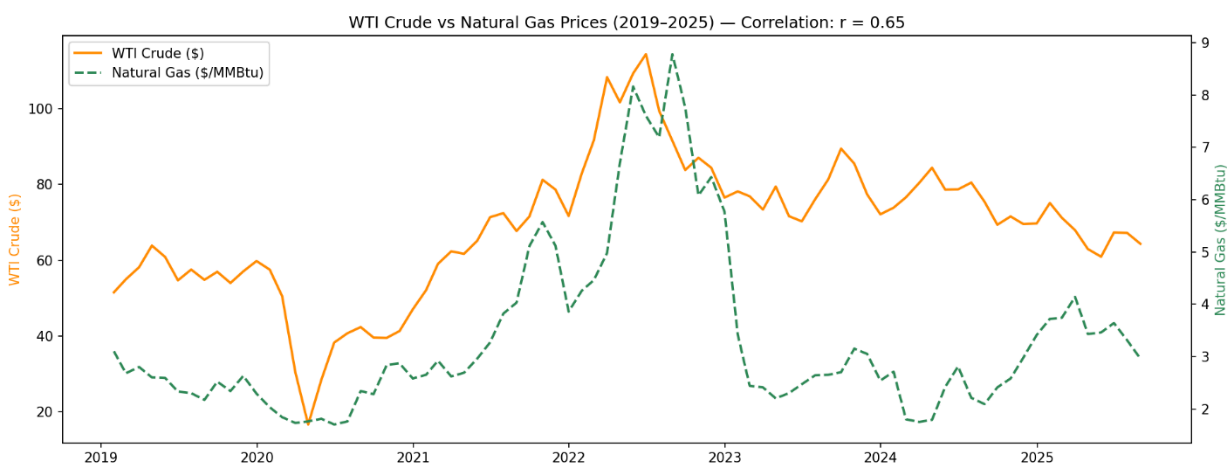
What the shaded regions mean:

- **Red shading (COVID-19 demand collapse, Feb–Jul 2020)** → Both lines drop simultaneously. Oil demand collapsed as lockdowns halted transportation worldwide. Tanker traffic also fell because less oil was being ordered globally
- **Orange shading (Ukraine war, Feb-Aug 2022)** → WTI spikes to its dataset high of around $114. Tanker activity through Hormuz remained relatively normal during this period
    - The disruption was primarily to Russian pipeline gas to Europe, not Gulf tanker routes. This is visible in the chart as the orange line spikes while the blue line stays relatively flat
- **Purple shading (Houthi disruptions, Nov 2023–Jan 2025)** → The most important region for this project. The blue dashed line drops while the orange solid line stays elevated. This is the divergence the rolling correlation chart captures numerically (fewer ships getting through, but prices not falling as a result)

How to interpret this chart:

- Before 2022, the two lines loosely tracked each other
    - When tanker activity was higher, oil prices tended to be higher, and vice versa
- The COVID shading shows the one period where both collapsed together
- The Houthi shading shows the opposite:
    - A supply shock where less oil moved through Hormuz, but prices did not fall, because global demand remained strong, and rerouting added costs
- Main takeaway: this is the storytelling chart. It connects the numerical findings from Visualizations 2 and 4 to the events that explain why the patterns changed

Visualization 6 →WTI Crude vs Natural Gas Prices (2019-2025)



This is another dual-axis time series chart, this time comparing two energy commodities against each other over the full 2019-2025 window to see whether they move together or independently.

Axes:

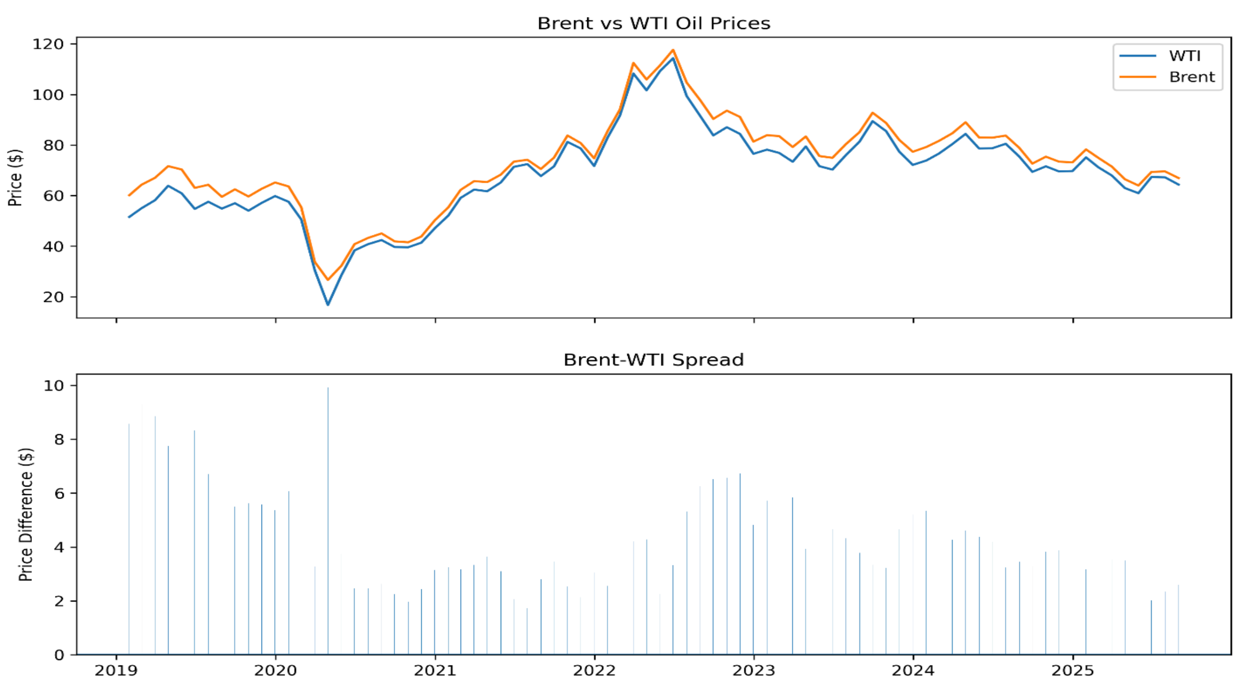
- **Left Y-Axis (orange)** ⟶ WTI Crude price in dollars per barrel
- **Right Y-Axis (green)** ⟶ Natural Gas price in dollars per MMBtu
    - The two y-axes are necessary because oil and gas are priced in completely different units at completely different scales:
        - oil around $20–115
        - gas around $1.50–9
- **X-Axis** ⟶ Time, January 2019 to August 2025

What the lines mean:

- **Solid orange line** ⟶ WTI crude oil price each month
- **Dashed green line** ⟶ Natural gas price each month
    - The dashed line for natural gas visually separates it from the WTI line, where they cross or run close together

Main takeaway: oil and natural gas share some common drivers (global energy demand, geopolitical shocks) but also operate independently. Natural gas has its own seasonal supply dynamics that crude oil does not, which explains why they diverge after 2022

Visualization 7 ⟶ WTI vs Brent Spread



This chart has two stacked panels showing two related but different things:

- The top panel compares the raw price levels of WTI and Brent directly
- The bottom panel shows the difference between them (the spread) as a bar chart

Top Panel Axes:

- **Y-Axis** → Price in dollars per barrel for both WTI and Brent
- **X-Axis** → Time, January 2019 to August 2025
- **Blue line** → WTI crude price
- **Orange line** → Brent crude price

Bottom Panel Axes:

- Y-Axis → Price difference in dollars (Brent minus WTI)
    - Positive bars mean Brent was more expensive than WTI that month (which is the normal state)
    - A bar of zero would mean they were priced identically
    - Negative bars would mean WTI was more expensive than Brent, which is unusual and would signal a specific regional supply dynamic
- **X-Axis** → Time, same range as the top panel
- **Blue + grey bars** → The spread value each month

What the colors mean:

- The bars in the bottom panel are a single color (blue/grey) because they represent one variable:
    - the size of the gap between the two prices

Main takeaway: WTI and Brent are effectively the same variable for the purpose of correlation analysis. However, the spread is worth tracking because it widens specifically during periods of global shipping disruption, making it a subtle secondary signal for the Hormuz analysis. When the Strait is under stress, Brent tends to price in that risk faster than WTI does

Conclusion:

Overall, inflation and oil prices show a meaningful relationship, but this relationship is not consistent over time.

In more stable periods, they tend to move together.
However, during global disruptions, the pattern changes.

Tanker traffic is not treated as a direct factor, but it helps provide context for understanding changes in global supply conditions.

In summary, oil prices and inflation are connected, but global events can alter how that relationship behaves.